CLOUD & DEVOPS

AWS Core Services Visual Guide

EC2, S3, VPC, IAM, RDS, Lambda —
the six primitives behind a typical 3-tier app.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Meet the Six Core Services

Each solves one job; together they run a business.

EC2
virtual servers

S3
object storage

VPC
private network

IAM
identity & access

RDS
managed databases

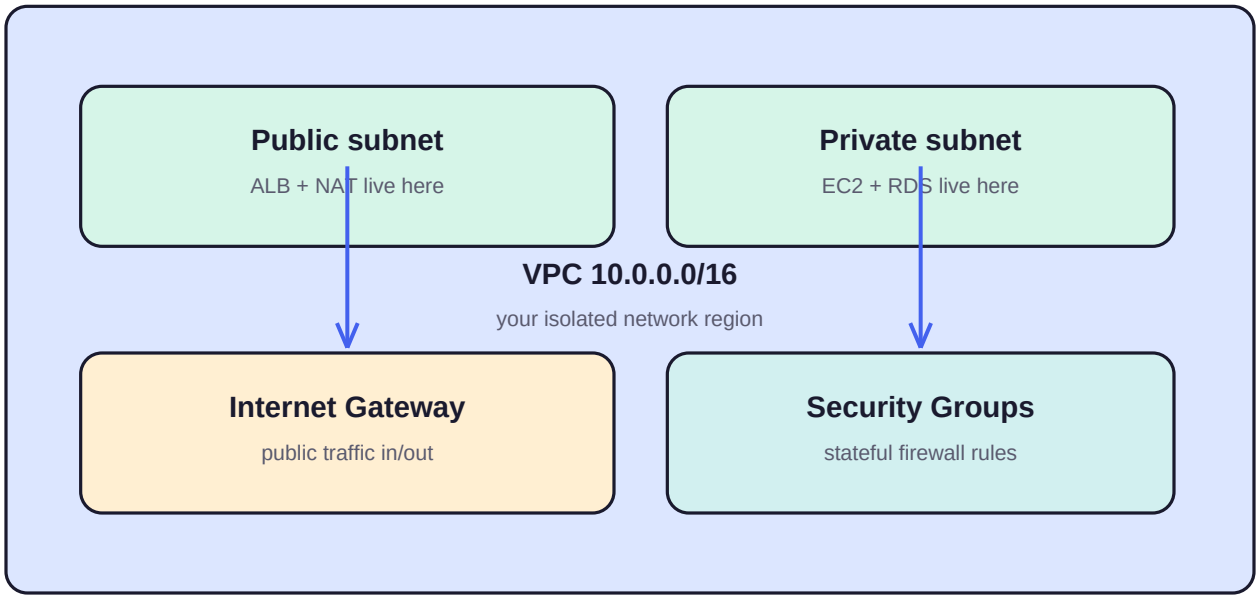
Lambda
serverless functions

Rule of thumb: compute (EC2/Lambda) runs code, S3/RDS hold data, VPC connects, IAM guards.

STEP 2/6

VPC: Your Private Network

Isolated network with public and private subnets.

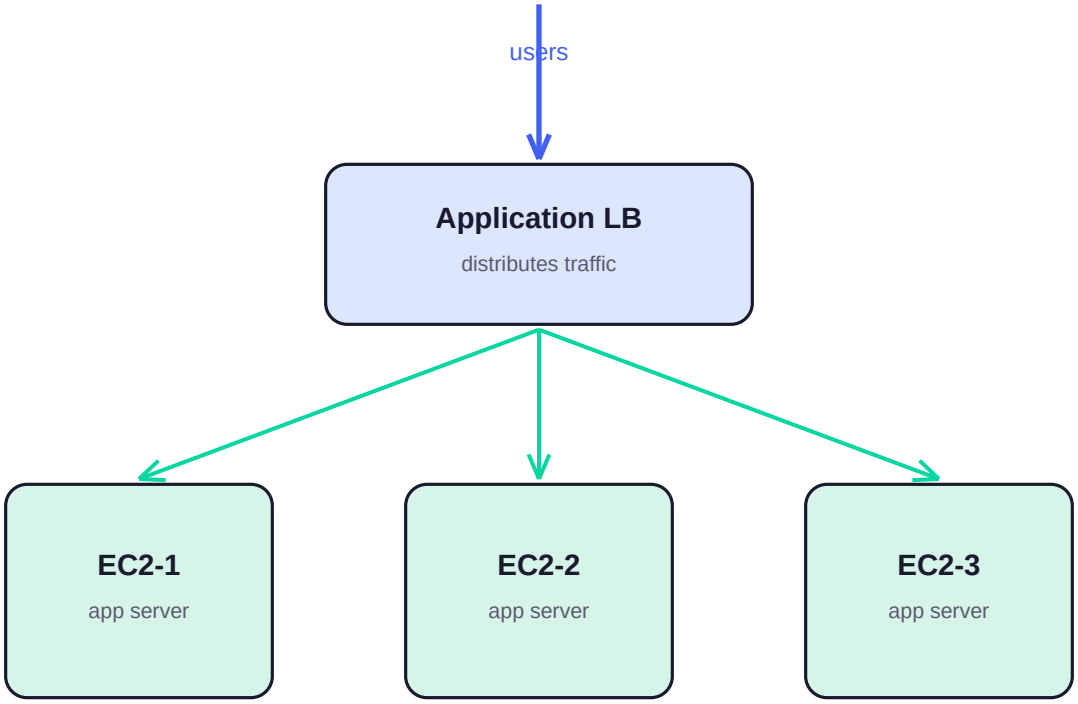


Private subnets reach the internet via NAT (outbound only) — databases never get public IPs.

STEP 3/6

The Web Tier: EC2 Behind a Load Balancer

Stateless app servers scale horizontally.



Auto Scaling group: add/remove EC2 instances on CPU or request-count metrics.

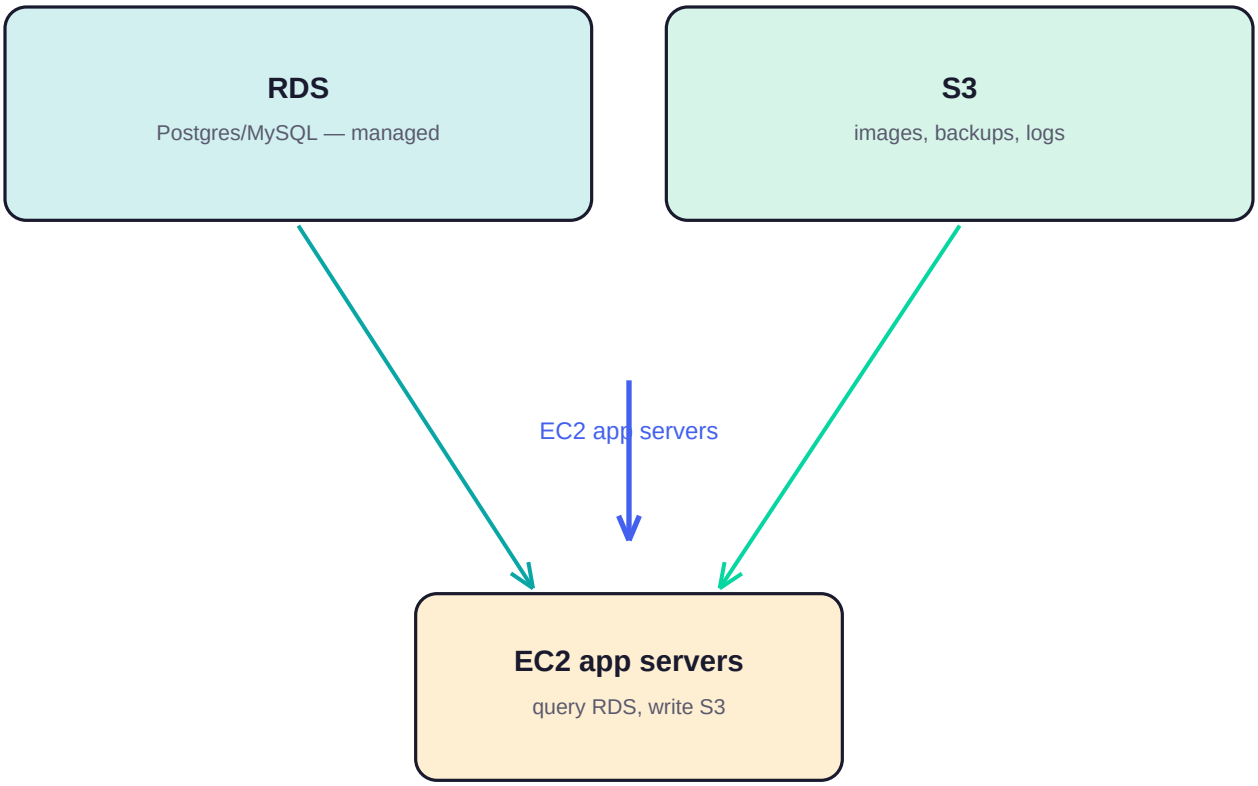
Health checks pull dead instances out of rotation automatically.

Keep app servers stateless: sessions in ElastiCache, uploads straight to S3.

STEP 4/6

The Data Tier: RDS + S3

Managed database for rows; object store for everything else.

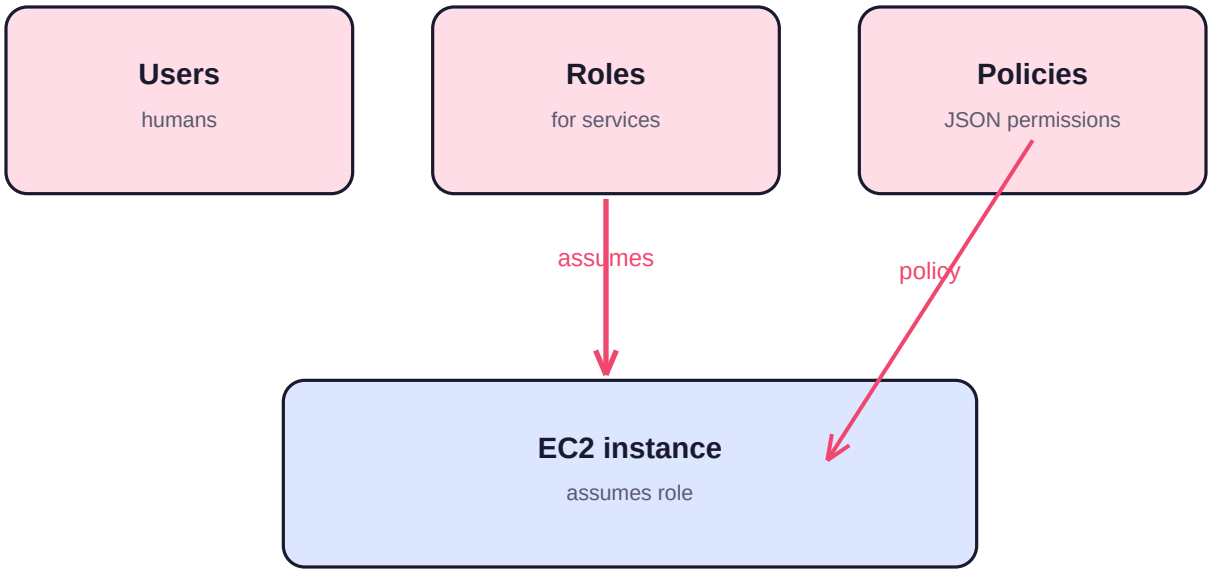


RDS Multi-AZ: synchronous standby in another zone — automatic failover, no data loss.

STEP 5/6

IAM: Who Can Do What

Identity first — every call is authenticated and authorized.



Least privilege: grant only the actions needed. EC2 gets an IAM role — never embed access keys.

STEP 6/6

Lambda: Events In, Code Out

Serverless glue for async work and automation.

Other triggers: API Gateway (REST APIs), SQS queues, DynamoDB streams, EventBridge schedules.

- No servers to manage — scales from 0 to thousands of concurrent executions.
- Pay per invocation + duration: ideal for spiky, event-driven work.
- 15-min max runtime: not for long jobs — that's EC2/ECS territory.

STEP ✓I✓

Key Takeaways

AWS Core Services — Remember

- VPC isolates your network: public subnets for load balancers, private for app and data.
- EC2 + Auto Scaling + ALB = the classic stateless web tier.
- RDS handles relational data (use Multi-AZ); S3 handles objects at any scale.
- IAM roles over keys; least-privilege policies on everything.
- Lambda glues events to actions — perfect for async, spiky workloads.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev